# Static Gesture Classification — Run Report

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## Data paths

| | |
|---|---|
| **Training data path 1** | /home/jestin/ThesisRepo/ML/NewTestData/Alan/Static · 35 trials |
| **Training data path 2** | /home/jestin/ThesisRepo/ML/NewTestData/Alex/Static · 35 trials |
| **Training data path 3** | /home/jestin/ThesisRepo/ML/NewTestData/Georgia/Static · 35 trials |
| **Training data path 4** | /home/jestin/ThesisRepo/ML/NewTestData/Henry/Static · 35 trials |
| **Training data path 5** | /home/jestin/ThesisRepo/ML/NewTestData/Jestin/Static · 36 trials |
| **Training data path 6** | /home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Static · 30 trials |
| **Training data path 7** | /home/jestin/ThesisRepo/ML/NewTestData/Josh/Static · 35 trials |
| **Training data path 8** | /home/jestin/ThesisRepo/ML/NewTestData/Marcus/Static · 35 trials |
| **Training data path 9** | /home/jestin/ThesisRepo/ML/NewTestData/Tash/Static · 0 trials |
| **Total training paths** | 9 · 276 trials total |
| **External test root** | /home/jestin/ThesisRepo/ML/NewTestData/Nat/Static |
| **Report output dir** | /home/jestin/ThesisRepo/ML/saved_reports |

## Sensor selection

| | |
|---|---|
| **Hands** | left, right |
| **Segments** | palm_mid, thumb, index, middle, ring, pinky, wrist |
| **Modalities** | YPR, Accel, Flex |
| **Sensor columns** | 176 channels |
| **Classes** | 7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman |

## Preprocessing, features & split

| | |
|---|---|
| **Resample steps** | 90 |
| **Butterworth** | on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz |
| **Normalisation** | standard |
| **Feature mode** | stats → feature dim 1408 |
| **Test size · seed · CV folds** | 0.2 · random_state=42 · 10-fold stratified |
| **Sample counts** | train (post-aug)=440 · test=56 |

## Augmentation (training only)

| | |
|---|---|
| **Enabled** | yes |
| **Copies per train trial** | 1 |
| **Jitter** | off · sigma=0.01 |
| **Amplitude scale** | on · range=(0.8, 1.2) |
| **Time warp** | off · range=(0.9, 1.1) |

## Classifiers — cross-validation & hold-out test

| Algorithm | CV mean (10-fold) | CV std | Hold-out test acc |
|---|---|---|---|
| Random Forest | 0.9955 | 0.0136 | 0.9643 |
| Voting Soft (SVM+RF+KNN+LogReg) | 0.9955 | 0.0136 | 0.9643 |
| Stacking (SVM+RF+KNN+LogReg) | 0.9955 | 0.0136 | 0.9643 |
| Logistic Regression | 0.9955 | 0.0136 | 0.9464 |
| SVM (Linear) | 0.9977 | 0.0068 | 0.9286 |
| SVM (RBF) | 0.9955 | 0.0136 | 0.8929 |
| KNN | 0.9273 | 0.0334 | 0.7857 |
| LDA | 0.9091 | 0.0419 | 0.1964 |

## Per-class hold-out F1

| Class | Random Forest | Voting (soft) | Stacking | Logistic Regression | SVM (Linear) | SVM (RBF) | KNN | LDA | Support |
|---|---|---|---|---|---|---|---|---|---|
| Double_Closed_Fist | 0.947 | 1.000 | 1.000 | 0.947 | 0.900 | 0.889 | 0.700 | 0.000 | 9 |
| Double_Nothing | 0.833 | 0.923 | 0.923 | 0.923 | 0.833 | 0.800 | 0.833 | 0.375 | 7 |
| Double_Okay | 1.000 | 0.933 | 0.933 | 0.933 | 0.933 | 0.857 | 0.625 | 0.174 | 8 |
| Double_Open_Palm | 1.000 | 0.941 | 0.941 | 0.941 | 0.941 | 0.941 | 0.778 | 0.348 | 8 |
| Double_Phone | 1.000 | 1.000 | 0.941 | 0.875 | 0.875 | 0.824 | 0.667 | 0.125 | 8 |
| Double_Pistol | 0.941 | 0.941 | 1.000 | 1.000 | 1.000 | 1.000 | 0.889 | 0.143 | 8 |
| Double_Spiderman | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.933 | 1.000 | 0.000 | 8 |
| **macro avg** | **0.960** | **0.963** | **0.963** | **0.946** | **0.926** | **0.892** | **0.785** | **0.166** | **56** |

## External test (NewTestData)

| Algorithm | External test acc |
|---|---|
| Random Forest | 0.9459 |

| | |
|---|---|
| Stacking (SVM+RF+KNN+LogReg) | 0.9459 |
| SVM (RBF) | 0.8378 |
| SVM (Linear) | 0.8378 |
| Voting Soft (SVM+RF+KNN+LogReg) | 0.8378 |
| Logistic Regression | 0.8108 |
| KNN | 0.7568 |
| LDA | 0.3243 |

### Per-class external accuracy

| Class | Random Forest | Stacking | SVM (RBF) | SVM (Linear) | Voting (soft) | Logistic Regression | KNN | LDA | Support |
|---|---|---|---|---|---|---|---|---|---|
| Double_Closed_Fist | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.000 | 6 |
| Double_Nothing | 0.600 | 0.800 | 0.400 | 0.600 | 0.400 | 0.400 | 0.200 | 0.400 | 5 |
| Double_Okay | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.800 | 5 |
| Double_Open_Palm | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.000 | 5 |
| Double_Phone | 1.000 | 0.800 | 0.400 | 0.200 | 0.400 | 0.200 | 0.000 | 0.200 | 5 |
| Double_Pistol | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.833 | 6 |
| Double_Spiderman | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.000 | 5 |